

King Singson

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EDUCATION

California State University, Los Angeles

Los Angeles, CA

Bachelor of Science in Electrical Engineering, Computer Engineering Focus

Expected May 2027

- **Relevant Coursework:** Computer Organization, Advanced Digital Design, Embedded System Programming, Control Systems, Signals & Systems, Data Communications & Networking, Circuit Analysis, Electronics, Machine Learning Principles

TECHNICAL SKILLS

Programming & Scripting: C/C++, Python, Verilog (RTL & Testbenches), MATLAB, MIPS Assembly

Hardware & Embedded Systems: ARM Cortex-M (TI LaunchPad), Microcontroller Interfacing (ADC, DAC, Timers, Interrupts), NEMA Stepper Motors, Motor Drivers, Sensors & DAQ Systems

Circuit Analysis & Electronics: Analog Circuit Design, Op-Amps, BJT & MOSFET Transistor Biasing, Two-Port Networks, AC/DC Power Analysis, Transient & Frequency Response

EDA, Robotics & CAD Tools: Autodesk Fusion 360, Robot Operating System (ROS), KiCad (Schematic & 2-Layer Layout), ModelSim, LTspice, Linux/Unix

Test & Measurement Equipment: Digital Storage Oscilloscopes, Logic Analyzers, Digital Multimeters (DMM), Function Generators, Benchtop DC Supplies, Soldering

Systems, Signals & Controls: Guidance, Navigation & Control (GNC), Computer Vision, Feedback Control, PID Tuning, State-Space, Fourier & Laplace Transforms, TCP/IP, IPv4/IPv6 Routing, FSM

PROJECTS

Senior Capstone: 4-Axis Motorized Microphone Stand | *Systems Lead, C++, NEMA Steppers* Aug. 2026 – Present

- Lead a cross-functional engineering team (EE, ME, CS) as systems architect, delegating subsystem tasks and defining electromechanical integration milestones.
- Direct the mechanical fabrication of a rigid 4-axis (X, Y, Z, Yaw) positioning rig designed to safely cantilever studio microphones with minimal acoustic resonance.
- Mitigate technical risks by evaluating structural payloads, sizing NEMA stepper motors, and specifying quiet motor driver electronics for live recording environments.
- Develop embedded control logic and a user-friendly Human Machine Interface (HMI) to accurately program, store, and recall complex 3D microphone placements.

Autonomous Robotic Navigation & Perception (AUV) | *ROS, Python, C++, Computer Vision* Fall 2026

- Implemented guidance, navigation, and control (GNC) algorithms within the Robot Operating System (ROS) environment to coordinate autonomous vehicle motion.
- Integrated computer vision pipelines for real-time feature extraction, obstacle recognition, and environmental perception.
- Designed finite-state machine (FSM) control architectures in Python and C++ to manage autonomous robotic operational modes and decision logic.

Custom Printed Circuit Board (PCB) Design & EDA | *KiCad, Linux, Hardware Verification* Fall 2026

- Designed schematic capture and routed a custom two-layer PCB layout in KiCad, selecting component footprints, routing netlists, and generating manufacturing Gerber files.
- Configured continuous ground copper pours and optimized power trace widths to ensure signal integrity and proper thermal relief.
- Utilized Linux terminal environments and EDA tools for automated design rule checks (DRC) and hardware design verification.